

TCES 430 Project Proposal

Julie Ann and Mahri

Tarot Card Reading Plan:

Phase 1: Core Tarot Reader

🧩 *Goal: Build a working 3 card tarot reader, with LCD output and multi button input*

🔧 Tasks

- [] **Select LCD** (16x2 or possibly 20x4 HD44780) and wire it to TM4C123GH6PM
- [] **Initialize LCD** in 4-bit mode and test basic text output
- [] **Wire buttons** to GPIO pins with pull-up/pull-down resistors
- [] **Configure interrupts** for button presses (e.g., draw, shuffle)
- [] **Implement card logic**: random selection, orientation, meaning
- [] **Display reading** on LCD across multiple lines
- [] **Add SysTick timer** for inactivity reset or delay handling

Demo-ready version with full reading flow

Potential Additions:

🔔 *Goal: Add sound cues for startup and card draw*

🔧 Tasks

- [] **Wire piezo buzzer** to PWM-capable GPIO (e.g.PB6)
- [] **Configure PWM** to generate tones (e.g., startup chime, draw beep)
- [] **Trigger buzzer** on startup and after draw interrupt

🤖 *Goal: Make the tarot reader mobile and interactive*

🔧 Tasks

- [] **Wire L298N motor driver** to TM4C123GH6PM and connect DC motors
- [] **Configure PWM** for motor speed control
- [] **Implement movement logic**: forward, stop, turn

- ☐ **Add ultrasonic sensor** to detect proximity and stop near user
- ☐ **Mount LCD on bot** and display reading after arrival

Interactive, mobile tarot delivery system

✦ *Goal: Add creative features if time allows*

Tasks

- ☐ **Name entry** via button interface
- ☐ **Sound sensor** for clap-triggered draw
- ☐ **RGB LEDs** for aura feedback
- ☐ **Bluetooth transmission** to phone or second MCU